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Extraction of Carvone from Spearmint Using the PIE Method

By

Josephine Mueller
B.S. North Central College, 2017

Capstone Project Report

Submitted in partial fulfillment of the requirements

For the Degree of Master of Science,

With a Major in Analytical Chemistry

Governors State University
University Park, Illinois 60484

2022

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December 2022

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Abstract

The Polyol Induced Extraction (PIE) method is to be used to extract and quantify carvone from the leaves of a spearmint plant. The PIE method uses polyol for the extraction. Polyol is a separating agent used to separate organic solvents from water. Glycerol is used in the PIE method because it binds very tightly to water which allows the organic solvent (acetonitrile in this case) to separate out and form an upper organic phase. This method is a patented extraction technique. It is used for the purpose of separating natural organic compounds from plants such as walnut, Kentucky coffee beans, or, in this case, spearmint. Experiments from other members of the research group have involved extracting and quantifying menthol (**1**, Figure 1) from spearmint plants (*Mentha spicata*) using the PIE method. After completing a calibration curve with caffeine (**2**, Figure 1) to test the PIE method, this research will involve the extraction and quantification of carvone (**3**, Figure 1) from spearmint. Carvone is considered an essential oil and is the most abundant essential oil found in the spearmint leaves. Due to this abundance, it is predicted that carvone in the spearmint leaves would be found in quantities between 0.82 and 1.34 percent. The research initially involves extracting the carvone from both dry and fresh leaves and running different extraction conditions to determine the best research route. The partition coefficient of carvone using the PIE method is determined using pure carvone and used to determine the effectiveness of the PIE method with carvone. This partition coefficient was found to be 120. By the end of the research, a quantification of carvone in the spearmint is determined. Once the carvone has been extracted, the data revealed a percent yield of 0.0244% in the fresh leaves and the partition coefficient from the fresh leaves could not be determined.

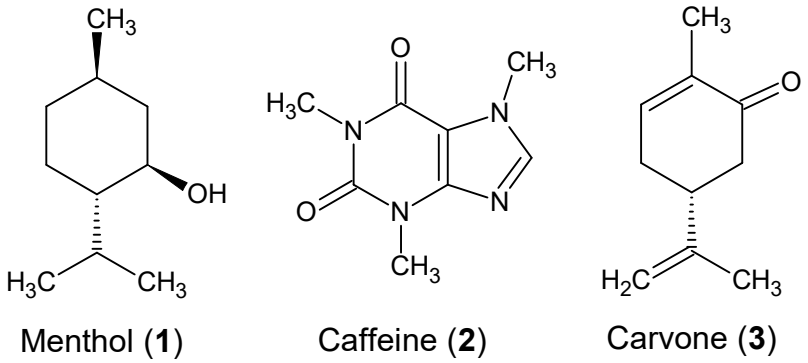

Figure 1. Chemical structures of menthol (**1**), caffeine (**2**) and carvone (**3**).

Keywords

PIE method, Carvone, Partition Coefficient, Calibration Curves, High Performance Liquid Chromatography (HPLC)

Introduction

Natural components found within essential oils like spearmint and other plants are useful for both the plant and for human use. These are typically very fragrant which aids in the identification of the plants. While not fully verified, it is believed different oils are essential for plant development and reproduction as well as protecting the plant from possible predators.[1] It is important to be able to isolate these essential oils from different plants for human use. The isolation should be done in the most efficient way possible. The spearmint plant has numerous essential oils present. Within the Sowa research lab, different essential oils are being extracted. This research is focusing on validating the Polyol Induced Extraction (PIE) method. It focuses on the ability of the PIE method to successfully extract carvone from the leaves.

There has been a growing interest in essential oils throughout the world for their wide variety of uses for both humans and animals. A study has been completed using plants from the Midwest comparing essential oils found in peppermint (*Mentha x piperita L.*), and two types of spearmint (*Mentha x gracilis Sole* and *Mentha spicata L.*).[2] GC data from that study showed the presence of carvone within the spearmint in quantities up to 70%.[2] This study is related to the research being presented in this research because it focused on carvone in spearmint which happened to be from the same region of the country. Weather, soil, heat, and other factors change based on the

---

[1] O'Shea, S. K.; Von Riesen, D. D.; Rossi. L. L. Isolation and Analysis of Essential Oils from Spices *J. Chem. Educ.* **2012**, 89, 665-668. https://pubs.acs.org/doi/10.1021/ed101141w.

[2] Wu, Z.; Tan, B.; Liu, Y.; Dunn, J.; Martorell Guerola, P.; Tortajada, M.; Cao, Z.; Ji, P. Chemical Composition and Antioxidant Properties of Essential Oils from Peppermint, Native Spearmint and Scotch Spearmint *Molecules* **2019**, 24 (15) 2825. https://www.mdpi.com/1420-3049/24/15/2825.

different regions in the country so it was important to find a study from the same region as the research being presented.[2] The study indicates carvone should be detected in the spearmint leaves if it can be extracted from the leaves. One main difference is the instrument usage between the study and the research, but they still would be predicted to share similar results.

The objectives of this research include the following:

- Determination of the partition coefficient of pure carvone with the average and standard deviation of multiple trials;
- Determination of percent yield of carvone in spearmint via the PIE method;
- Determination of the partition coefficient of raw material (wet spearmint leaves).

Common extraction techniques such as steam/hydro distillation, liquid-liquid extraction, etc. use solvents that can be harmful to the environment. The PIE method is potentially an alternative extraction method yielding a higher purity in a more cost-effective manner.[3] The PIE method was initially developed as a way to separate acetonitrile from water to counter an acetonitrile shortage.[4] It was only recently tested for its use with natural products in order to separate essential oils which can then be used in industry. Research previously completed by Dr. Sowa and his research group

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[3] DelMastro, T. Polyol induced partitioning of essential oils in aqueous organic solvent mixtures. Ph.D. Dissertation. Seton Hall University, South Orange, NJ, 2016. http://proxy.govst.edu:2048/login?url=https://www.proquest.com/dissertations-theses/polyol-induced-partitioning-essential-oils/docview/1867775536/se-2 (accessed 2022-10-20).

[4] DelMastro, T.; Snow, N. H.; Murphy W. R.; Sowa, J. R. Polyol-induced partitioning of essential oils in water/acetonitrile solvent mixtures *J. Liq. Chromatogr. Relat. Technol.* **2017**, 40(7), 376-383. https://www.tandfonline.com/doi/full/10.1080/10826076.2017.1308379.

focused on optimizing the PIE method and many of the techniques were applied to this research. The partition coefficient was optimized at lower temperatures down to -20°C before the samples freeze which is like the temperature goal for this research.[4] The Sowa research found the carvone to be highly recoverable which was used to create some of the objectives for this experiment.

Carvone is the chosen essential oil for the extractions in this project. The PIE method should be very effective at isolating carvone and a high yield is expected from the spearmint leaves. Carvone has properties that have been proven beneficial in the pharmaceutical world.[5] These include antibacterial, antifungal, antioxidant, and anti-tumor properties as well as many others.[2] The pharmacological activities are important because there are currently numerous diseases with no known cure, but the ketone monoterpenes commonly found in plants are often able to be used in modern drugs and agriculture. Carvone (**3**) is a ketone monoterpene and relatively abundant within the spearmint.[5] According to Pina *et al*. Carvone has effects on both the nervous system and the hepatic metabolism and will hopefully show more promising effects on existing rare diseases.[5] As an anti-tumor agent, carvone's system produced enzyme-inducing activity which reduces tumor formation significantly.[5] Carvone is also used in insecticides as a repellent.[6] The benefits of carvone are numerous and it is important to efficiently extract it. Luckily, it is abundant in spearmint and has been proven to extract

---

[5] Pina, L. T. S.; Serafini, R.; Oliveira, M. A.; Sampaio, L. A.; Guimaraes, J. O.; Guimarae, A. G. Carvone and its pharmacological activities: A systematic review *Phytochem.* **2022**,196, 113080. https://doi.org/10.1016/j.phytochem.2021.113080.

[6] de Carvalho, C. C. R.; da Fonseca, M. M. R. Carvone: Why and how should one bother to produce this terpene *Food Chem.* **2006**, 95 (3) 413-422. https://doi.org/10.1016/j.foodchem.2005.01.003.

out. The test here will be whether or not carvone can be extracted from spearmint, but rather, whether it can be extracted specifically by the PIE method.

The different enantiomers of carvone have different qualities which means it is possible they would react differently to the PIE method. *"For example, the odor threshold value of (R)-(-)-carvone (0.043 ppm) is about ten times lower than that of its (S)-(+) enantiomer (0.60 ppm)."*[7] Sensory differences like this are seen in a variety of compounds which means overtime, the PIE method should be tested on the different enantiomers in order to fully understand the scope of the extraction success. According to Dr. Sowa, the HPLC methods used for this research and at Governors State University are unable to accurately distinguish between *R* and *S* carvone. While studies like the one completed by Ikan show there are differences, this project is limited by the abilities of the method and the instrument at Governors State University.[7] The PIE method products can be tested by methods other than HPLC if the enantiomer proves to be important. According to studies by *Ikan 2008*, when run on GC, the (*R*) enantiomer almost always elutes off the column before the (*S*) enantiomer.[7] As seen below in the method section, the pure carvone used in the methods section is the (*S*)-(+) enantiomer. The (*R*)-(-)-carvone is most commonly found in the spearmint.[8] The HPLC system used detects all carvone, so the enantiomer used for the determination of the partition coefficient should not matter. If any enantiomer of carvone is present in the spearmint, it will show up in the HPLC data and will not be distinguishable from other enantiomers during this study. If it was imperative to know which enantiomer was extracted, a TLC

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[7] *Selected Topics in the Chemistry of Natural Products, R. Ikan, Ed.* World Scientific, **2008**.
[8] Souza, F. V. M.; da Rocha, M. B.; de Souza, D. P.; Marcal, R. M. Carvone: Antispasmodic effect and mode of action *Fitoterapia* **2013,** 85, 20-24. https://doi.org/10.1016/j.fitote.2012.10.012.

plate could be used with both enantiomers spotted on the plate to determine which was present in the product. According to *Ikan 2008*, the naturally occurring carvone in spearmint is the (R)-(-)-carvone which should be more detectable than the enantiomer.

This experiment is using the wet leaves from freshly picked spearmint. Dried leaves would have been ideal because water is a major component in the leaves which causes the percent yield to seem very small, but the dried leaves required such a large volume of leaves to have enough for the extraction that fresh leaves were chosen as an alternative and they were easier to work with. Dr. Sowa measured the dried mass weight of spearmint leaves to determine the moisture content which was then used as a correction factor for the final percent yield values. A variety of experiments were performed to determine the optimal experimental conditions and numerous trials were ran to ensure reproducibility. This research group is testing the PIE method for its ability to separate natural compounds into acetonitrile from different types of plants. If the method works, the essential oils should be in the top layers of the extraction with the acetonitrile and can be identified using HPLC analysis.

Materials and Methods

HPLC instrument: The HPLC instrument was an Agilent 1260 Infinity HPLC System with a DAD detector. The instrument was operated with the Agilent OpenLab CDS ChemStation Edition for LC and LC/MS Systems software Rev. C.01.05 [35], copyright 2001-2013. Absorbance values will be reported to two decimal places.

The HPLC analysis of caffeine was performed using the “caffeine baldrich sowa” method with a C18 column. The flow rate was set to 2 mL/min using 78% an aqueous solution of 0.1% formic acid in water and 22% acetonitrile.

The column was changed after analysis of caffeine and the HPLC analysis of carvone was performed using a Grace Platinum C18-EPS column (3 µm, 53 mm, ID 7mm) with lot # 61/010, part # 50573, and serial # 615060461. The method used was the “Carvone60MeOH 0.5mL_min2022MR10” method with the following parameters: 0.5 mL/min flow rate, 60% MeOH and 40% H_2O, injection volume 1 µL, DAD 254 nm, and a column temperature of 25°C.

A portion of this experiment was completed using pure carvone to test the method. The pure carvone used was (*S*)-(+)-Carvone. The CAS-No is 2244-16-8 made by the Millipore Corporation. Other materials used in the experiments were caffeine, 0.1% formic acid in water, glycerol, acetonitrile, and HPLC grade water all purchased from VWR, Inc.

The beginning of experimentation was done by creating a caffeine calibration curve and running the PIE extraction using caffeine. This was a way to learn the PIE

method and for the group to begin collecting a lot of data on caffeine and the PIE method. Each member of the research group starts with caffeine so over time a huge data set will be collected. The method was chosen based on PIE method research which added glycerol and allowed the samples to cool down before analysis.[9] Table 1 shows the data used to create the caffeine calibration curve which can be found in Figure 2. Figure 3 shows the HPLC results of one of the chromatograms from the calibration. The slope found from the curve can then be used in calculations to determine the concentrations from different experiments.

Table 1. Caffeine Calibration Data

Concentration (mg/mL)	Peak Area (mAU*s)
1.0	1757.34
2.5	3991.77
5.0	8450.12
10	12631.57
15	15574.41

[9] Patel, S.; Snow N. H. Extraction of glucocorticoids from water into acetonitrile using polyol induced extraction with ultra performance liquid chromatography and triple quadrupole mass spectrometry (PIE-UPLC-MS-MS) *J. Liq. Chromatogr. Relat. Technol*. **2022**. Published online, https://doi.org/10.1080/10826076.2022.2110116 (accessed 2022-10-20).

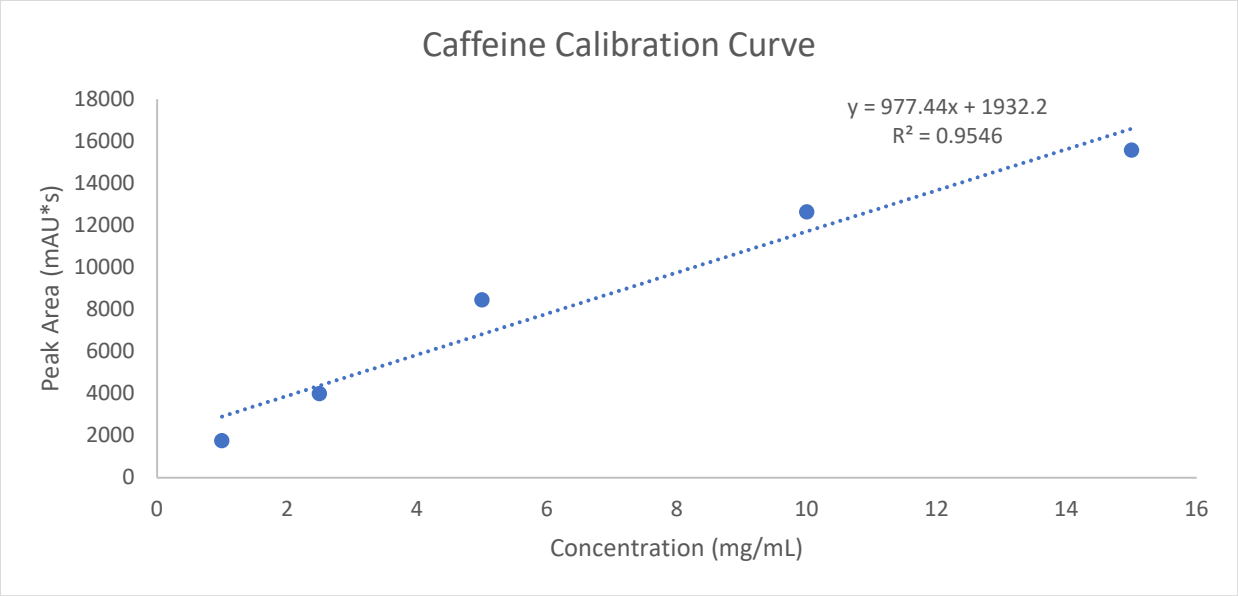

Figure 2. Caffeine Calibration Curve

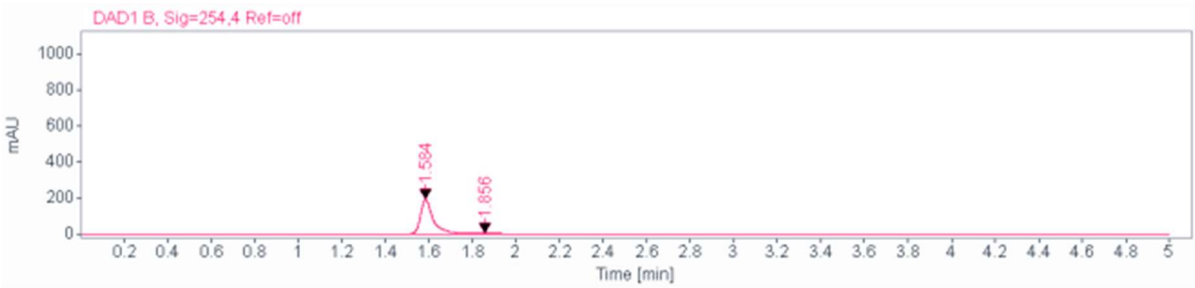

Figure 3. Chromatogram from 1.0 mg/mL Caffeine Calibration Standard

Once the calibration was completed, the next step was identifying the percent recovery and partition coefficient of caffeine using the PIE method. On one centrifuge tune was added 2.0 g of glycerol. In a separate 15 mL centrifuge tube, 70 mg of caffeine was added with a spatula. Then a 70:30 mixture of acetonitrile and HPLC grade water was added until the total volume in the centrifuge tube was at 10.0 mL. The mixture was then inverted until the caffeine dissolved. Once thoroughly mixed, this mixture was added to the centrifuge tube containing the glycerol and once again mixed by inversion.

The centrifuge tube was then inserted in the freezer where it was allowed to equilibrate for about one hour until it reached -18 °C.

After the vials reached the desired temperature, there was a visible separation of layers within the centrifuge tube. If the partition coefficient is high enough, the caffeine should have a higher affinity for the acetonitrile in the top layer while the bottom layer should have the glycerol and water. A high partition coefficient would confirm the ability to use the PIE method for caffeine extractions. A one milliliter aliquot was taken from the top layer and from the bottom layer separately to be run on the HPLC. For caffeine, a second aliquot was taken from each layer for comparison. The peak area was then used in calculations of the partition coefficient.

The three objectives of this research were all reached using the PIE method and a carvone calibration curve. Initially, pure carvone was used to create a calibration curve and the linear equation determined from the curve was then used in subsequent experiments to determine partition coefficients. Three data points were used to create the carvone calibration curve. After using five data points for caffeine, it was determined only three were needed for future curves. Concentrations of 0.10 mg/mL, 0.75 mg/mL, and 1.5 mg/mL were created using pure liquid carvone and a 70:30 mixture of acetonitrile and HPLC grade water. When run on the HPLC using the “Carvone60MeOH 0.5mL_min2022MR10” method, results for the calibration curve were created and a slope for the calibration line was found which was then used throughout the other experiments. Table 2 shows the data used to create the calibration curve shown in

Figure 4. Figure 5 shows the HPLC chromatogram used for the peak area of the 0.10 mg/mL standard of pure carvone for the calibration curve.

Table 2. Carvone Calibration Data

Concentration (mg/mL)	Peak Area at 254 nm (mAU*s)
0.1	419.13
0.75	2781.89
1.5	5862.00

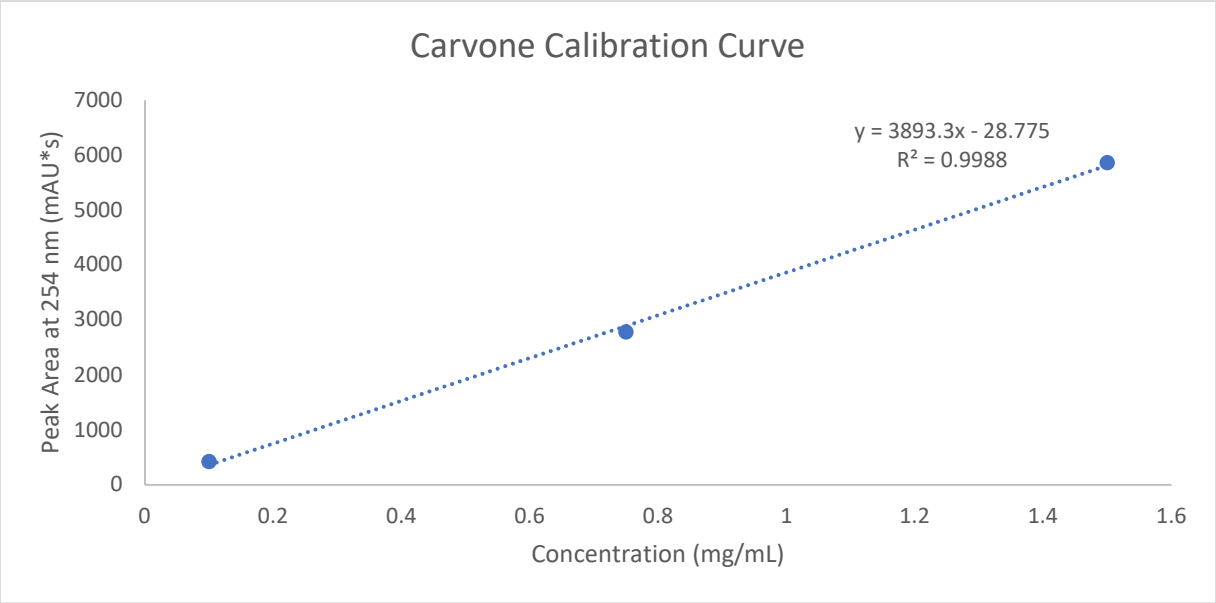

Figure 4. Carvone Calibration Curve

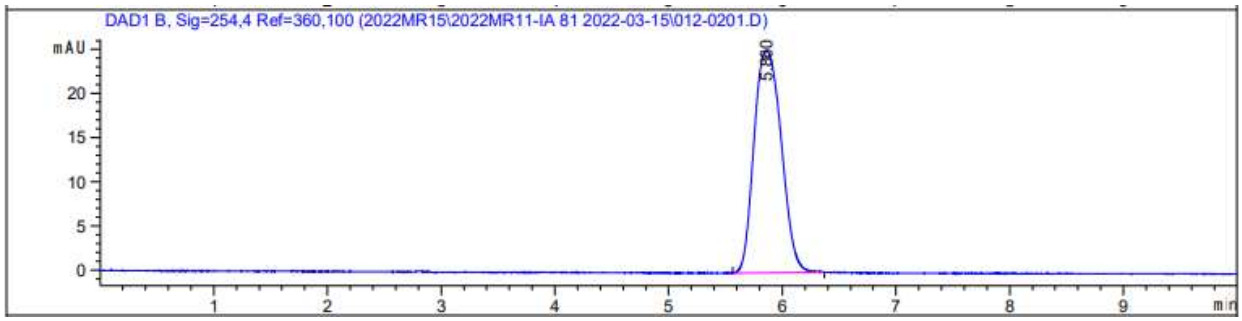

Figure 5. Chromatogram from 0.10 mg/mL Carvone Calibration Standard

Once the calibration was completed, the next step was identifying the partition coefficient of pure carvone using the PIE method. This method was very similar to the caffeine method above with some minor differences. Glycerol (2.0 g) were added to a 15 mL centrifuge tube. In a separate 15 mL centrifuge tube, a pipette was used to add one gram of pure carvone. Then a 50:50 mixture of acetonitrile and HPLC grade water was added until the total volume in the centrifuge tube was at 10.0 mL and the solution was mixed. This mixture was added to the centrifuge tube containing the glycerol and once again mixed by inversion. The centrifuge was then inserted in the freezer. A new, more accurate thermometer had been added to the freezer at this point. Instead of the original -18°C like caffeine, the carvone was allowed to equilibrate to -20 ^{o}C for one hour now that the thermometer gave a clearer reading.

After this hour, there was a visible separation of layers within the centrifuge tube. If the partition coefficient is high enough (which is the prediction), the carvone should have a higher affinity for the acetonitrile in the top layer while the bottom layer should have the glycerol and water. If a high partition coefficient is seen, it would confirm the ability to separate carvone and quantify it allowing for a more accurate determination of the carvone present in spearmint leaves. A one milliliter aliquot was taken from the top layer and from the bottom layer separately to be run on the HPLC. After the initial run, it was determined the top layer samples were highly concentrated and needed to be diluted. A 1:100 dilution was performed using the 50:50 acetonitrile:water mixture and these samples were then run on the HPLC. This dilution worked and was used for all top layer samples from the pure carvone experiments. Using the method from above,

the peak area is then used in calculations of the partition coefficient. The schematic below in Figure 6 demonstrates the process visually.

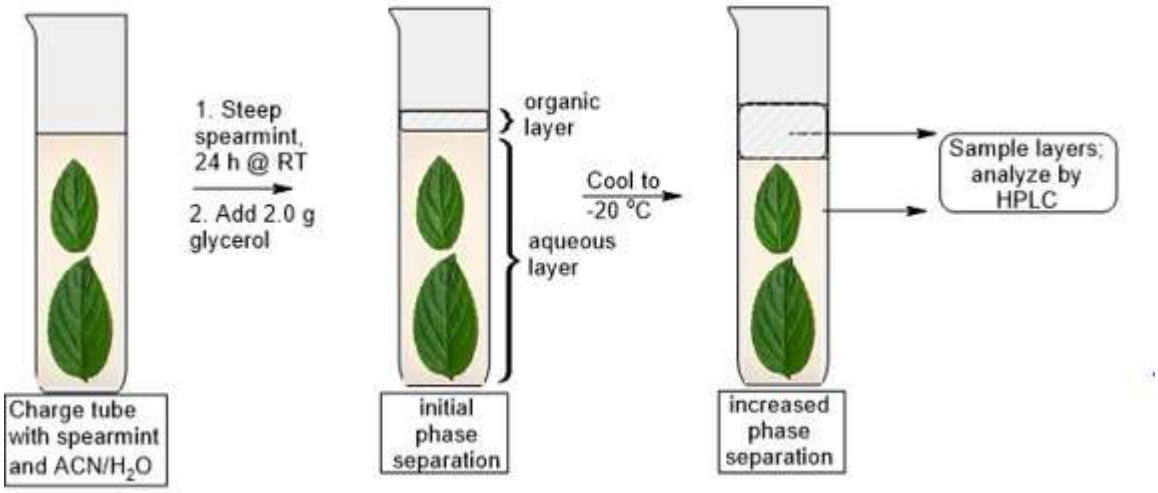

Figure 6. Schematic Showing the PIE Method with Leaves

A percent recovery of carvone was determined for each top layer and then a partition coefficient was determined for each trial using both the top and bottom layer results. An average partition coefficient was calculated from each experiment and an overall average was determined after the three pure carvone experiments which used a total of 9 different trials and partition coefficients. From the HPLC data, the peak area of the top layer is first used to calculate the concentration by taking the peak area and dividing that by the slope determined in the carvone calibration curve. Once this concentration was determined, it was multiplied by 100 to account for the dilution made. Then, the amount of carvone present in the top layer was determined by taking the volume of liquid in the top layer and dividing it by the adjusted concentration. In order to determine percent recovery, the amount of carvone in the top layer was divided by the

starting weight of the carvone added to the tube and this whole value was multiplied by 100.

The below calculations were used to determine the percent recoveries and partition coefficients of all the trials throughout the research.

Example Calculations:

$\left(\frac{Peak\ Area\ from\ Top\ Layer}{Slope\ from\ Calibration\ Curve}\right) * 100 =$ Concentration of Top Layer

$$\frac{Volume\ of\ Top\ Layer}{Concentration\ of\ Top\ Layer} = Amount\ of\ Carvone\ in\ Top\ Layer$$

$$\left(\frac{Amount\ of\ Carone\ in\ Top\ Layer}{Starting\ Weight\ of\ Carvone}\right) * 100 = Percent\ Recovery$$

$\left(\frac{Peak\ Area\ from\ Bottom\ Layer}{Slope\ from\ Calibration\ Curve}\right) =$ Concentration of Bottom Layer

$$\frac{Volume\ of\ Bottom\ Layer}{Concentration\ of\ Bottom\ Layer} = Amount\ of\ Carvone\ in\ Bottom\ Layer$$

$\frac{Concentration\ of\ Top\ Layer}{Concentration\ of\ Bottom\ Layer} =$ Partition Coefficient

The bottom layer followed the same calculations to get the amount of carvone present in the layer using the peak area from the bottom layers. The concentration did not need to be adjusted for dilutions this time. Once the amount of carvone was determined for both the top and bottom layers, the mass balance can be calculated. To do this, the amount of carvone in the top layer was divided by the amount of carvone in the bottom layer. The final step was determining the partition coefficient. The adjusted concentration from the top layer was divided by the concentration from the bottom layer which gave the partition coefficient.

A total of five different sets of experiments were run and each experiment was run in triplicate. Only the last three were used to determine the average partition coefficient of pure carvone. The first two weeks of experiments were used to learn the PIE method and practice using the HPLC. It was decided by the third week, the researcher was proficient enough in the method for the data to be considered reliable. Experiments 3, 4, and 5 were completed and the results were used to determine an average partition coefficient which showed carvone had a high affinity for the acetonitrile.

Once the partition coefficient was determined for pure carvone, the next goal was to determine the partition coefficient and total recovery of carvone from spearmint leaves. Experiments were set up to determine the optimal conditions needed to pull the essential oils out of the leaves and into solution. Both fresh/wet and dried leaves were attempted and the fresh/wet leaves were determined to be the best option for these experiments. The dried leaves required far too many fresh leaves to be dried, crushed, and sifted in order to collect enough pieces to reach one gram for every experiment. Then, the time needed for steeping had to be determined. Once the fresh leaves were submerged in the acetonitrile/water mixture, they needed to steep for a set amount of time in order for the oils to extract from the leaves. Then they needed to equilibrate in the freezer for a set time as well to keep the experiments consistent. Previous researchers determined the PIE method worked best if the mixture was allowed to reach -20 °C before the aliquots were taken for HPLC analysis. The experiment options were a 30-minute steep with a 24-hour equilibrate time, a 24-hour steep with a 24-hour equilibrate time, and a 24-hour steep with a one-hour equilibrate time. No carvone was

detected from the 30-minute steep time so a 24-hour steep time was determined to be the best condition. There did not appear to be any difference between equilibrating for one hour or 24-hours so the one-hour timeframe was the chosen method due to timing working out better for the overall experiment.

With the conditions determined, two experiments with spearmint leaves were set up in different weeks, each in triplicate, for a total of six measurements from the fresh leaves. A 15 mL centrifuge tube had one gram of fresh spearmint leaves added. This was between six and ten leaves depending on their size. The leaves were rolled gently and placed in the tube so that liquid could still get around them and in the center to allow the full leaf contact with the solvent for steeping. This tube was then filled with 10.0 mL of the 50:50 acetonitrile:water mixture. The tube was inverted multiple times and then allowed to steep for 24 hours at room temperature in the laboratory. After steeping for 24 hours, glycerol (2.0 g) was added to the centrifuge tube, the tube was inverted, and then placed in the freezer. The mixture was allowed to equilibrate in the freezer for one hour at which point it had reached -20 °C. This was determined by a probe inserted into a blank sample next to the test samples connected to a digital thermometer. The tube was then taken out of the freezer, where a separation of top and bottom layers could once again be seen, although it is harder now because the fresh leaves sometimes were blocking the view. When the aliquots were taken, it was easy to verify the top and bottom layers though because the top layer in every trial were a dark green liquid and the bottom layer in every trial was a dark brown liquid. The one milliliter aliquots were drawn up into a syringe and then filtered using an EZFlow® Syringe Filter with a Foxx Hydrophilic PVDF Membrane. The filters had a 13 mm diameter and a pore

size of 0.22 μm. The liquid was filtered directly into an HPLC vial and then ran using the carvone method described above. Once the HPLC runs were complete, the average partition coefficient and the total yield of carvone in the spearmint leaves was determined. These used the same calculations as the pure carvone experiments, minus the dilution factor.

Results and Discussion

A total of three trials were completed for the caffeine PIE method tests, and two aliquots were taken from each of the top and bottom layers to see if there was any difference between them. Table 3 gives all the calculations from the top layers and Table 4 shows all the bottom layer calculations. Figures 7 and 8 give one of the chromatograms for a top layer and a bottom layer of the caffeine analyzed on the HPLC. There was a decent percent recovery seen for all the trials. The method seemed effective enough in recovering the caffeine to give usable and reliable data. The partition coefficients were all very small (<1) which means the caffeine is much more soluble in water than it is in the acetonitrile. This would mean the PIE method would be ineffective on caffeine extractions according to this data.

Table 3. Caffeine Top Layer Calculations

Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Top Volume (mL)	Amount of Caffeine in Top (mg)	Starting mg	% Recovery
Top 1	7987.39	8.17	3.00	24.5	71.5	34
Top 2	8745.39	8.95	3.50	31.3	71.2	44
Top 3	8316.38	8.51	3.40	28.9	70.8	41
Top 1'	8040.84	8.23	3.00	24.7	71.5	35
Top 2'	8506.77	8.70	3.50	30.5	71.2	43
Top 3'	8220.17	8.41	3.40	28.6	70.8	40

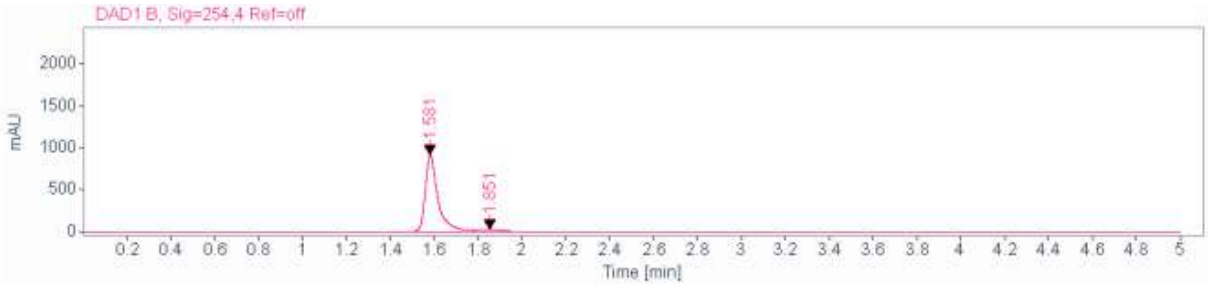

Figure 7. Chromatogram from Caffeine Top Layer

Table 4. Caffeine Bottom Layer Calculations

Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Bottom Volume (mL)	Amount of Caffeine in Bottom (mg)	Mass Balance (mg)	Partition Coefficient
Bottom 1	8467.73	8.66	8.25	71.5	96.0	0.94
Bottom 2	9043.15	9.25	8.25	76.3	108	0.97
Bottom 3	9005.75	9.21	8.40	77.4	106	0.92
Bottom 1'	8400.50	8.59	8.25	70.9	96	0.96
Bottom 2'	9088.61	9.30	8.25	76.7	107	0.94
Bottom 3'	9053.87	9.26	8.40	77.8	106	0.91

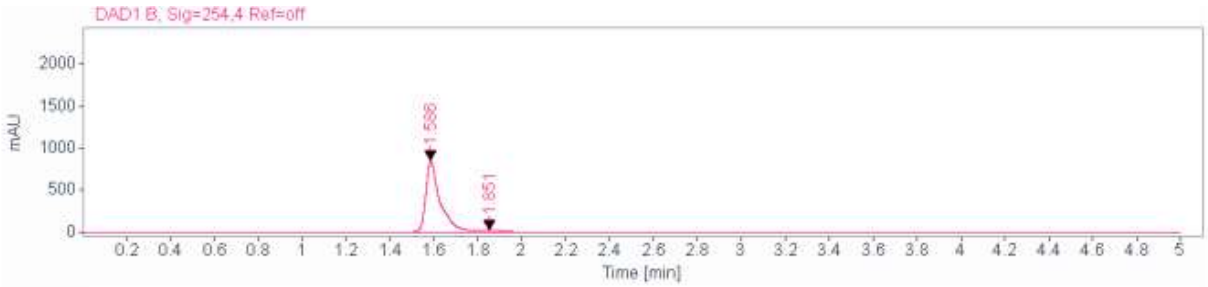

Figure 8. Chromatogram from Caffeine Bottom Layer

A total of five experiments were completed using pure carvone. The first two experiments were used as practice to make sure the method was understood. These two experiments yielded partition coefficients that were very different from one another. The partition coefficient of experiment 1 was only 64 while the partition coefficient of experiment 2 was 210. The results were not included in the calculation of the final partition coefficient value because these experiments were considered preliminary. After these two were completed, it was decided that enough experiments would be run until there were three in a row that showed consistent results and experiments 3-5 all worked well so only 3 more experiments were needed. There were similarities throughout the last three experiments not only in the final partition coefficient, but also in the values of all the calculations.

The top volume from each experiment and trial were all very consistent with one another. Table 5 shows all three trials had a top layer volume of 4.70 mL, Table 6 shows all three trials were between 4.60 mL and 4.90 mL, and Table 7 shows all three trials once again had a top layer volume of 4.70 mL. The consistency in these top layers was proof that all the experiments were consistent with one another. Tables 5, 6, and 7 show the calculations for the top layers of the nine trials. These contain the percent recovery of carvone from the PIE method as well. Experiment 3 had an average of 88% recovery, experiment 4 had an average of 90% recovery, and experiment 5 had an average of 78% recovery. The standard deviation between these experiments was 6.20. This is a low standard deviation which means the values are not dispersed very far from the mean. Overall, the average percent recovery from all nine trials was 85%. Figure 9 gives one of the chromatograms for a top layer of the pure carvone analyzed on the HPLC.

Table 5. Experiment 3 Top Layer Calculations

Experiment 3 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Adjusted Conc. For dilution	Top Volume (mL)	Amount of Carvone in Top (mg)	Starting mg	% Recovery
Top 1	7364.41	1.89	189	4.70	889	1008.0	88
Top 2	7354.75	1.89	189	4.70	888	1011.7	88
Top 3	7296.31	1.87	187	4.70	881	1002.9	88

Table 6. Experiment 4 Top Layer Calculations

Experiment 4 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Adjusted Conc. For dilution	Top Volume (mL)	Amount of Carvone in Top (mg)	Starting mg	% Recovery
Top 1	7387.51	1.90	190	4.90	9230	1008.0	92
Top 2	7584.80	1.95	195	4.70	916	1011.7	91
Top 3	7366.35	1.89	189	4.60	870	1002.9	87

Table 7. Experiment 5 Top Layer Calculations

Experiment 5 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Adjusted Conc. For dilution	Top Volume (mL)	Amount of Carvone in Top (mg)	Starting mg	% Recovery
Top 1	5983.86	1.54	154	4.70	722	1001.6	72
Top 2	7439.63	1.91	191	4.70	898	1041.1	86
Top 3	6416.28	1.65	165	4.70	774	1013.5	76

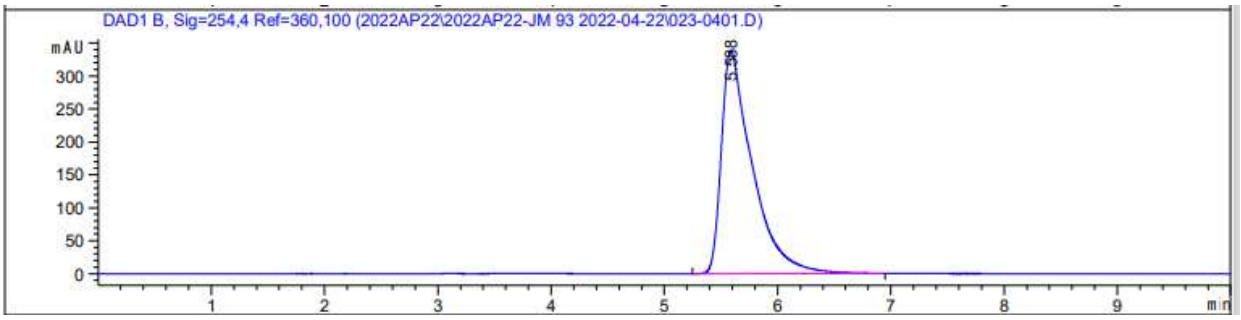

Figure 9. Chromatogram from Pure Carvone Top Layer

Once the top layers were analyzed, the bottom layers were analyzed. In order to calculate the partition coefficient, the concentration of both the top and bottom layer are needed. Tables 8, 9, and 10 show the values and calculations from the bottom layers of each experiment. Experiment 3 ended up with an average partition coefficient of 120, experiment 4 had an average partition coefficient of 120, and experiment 5 had an average partition coefficient of 110. The standard deviation between the three experiments was 4.60. The overall partition coefficient based on the average of all nine trials was 120 with a standard deviation of 16.5. The standard deviation showed very little dispersion between the different experiments. The high partition coefficient indicates carvone was much more soluble in the acetonitrile solvent and very insoluble in water making the PIE method an effective extraction technique based on the pure carvone data. The three experiments were consistent enough with a low enough standard deviation that it was determined no further experiments were needed. The

high percent recovery and the large, consistent partition coefficient confirmed the PIE method was an effective method for extracting carvone and could be used on the fresh spearmint leaves. Figure 10 gives one of the chromatograms for a top layer of the pure carvone analyzed on the HPLC.

Table 8. Experiment 3 Bottom Layer Calculations and Partition Coefficient

Experiment 3 Trial #	Peak Area (mAU *s)	Concentration (mg/mL)	Bottom Volume (mL)	Amount of Carvone in Bottom (mg)	Mass Balance (mg)	Partition Coefficient
Bottom 1	7584.54	1.95	8.80	17.1	906	97
Bottom 2	6089.00	1.56	8.80	13.8	902	120
Bottom 3	5641.04	1.45	8.80	12.8	894	130

Table 9. Experiment 4 Bottom Layer Calculations and Partition Coefficient

Experiment 4 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Bottom Volume (mL)	Amount of Carvone in Bottom (mg)	Mass Balance (mg)	Partition Coefficient
Bottom 1	5417.82	1.39	8.90	12.4	942	140
Bottom 2	5852.99	1.50	8.80	13.23	929	130
Bottom 3	7230.71	1.86	8.90	16.53	887	100

Table 10: Experiment 5 Bottom Layer Calculations and Partition Coefficient

Experiment 5 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Bottom Volume (mL)	Amount of Carvone in Bottom (mg)	Mass Balance (mg)	Partition Coefficient
Bottom 1	6571.60	1.69	8.90	15.0	737	91
Bottom 2	5709.44	1.47	8.80	12.9	911	130
Bottom 3	5337.42	1.37	8.90	12.2	787	120

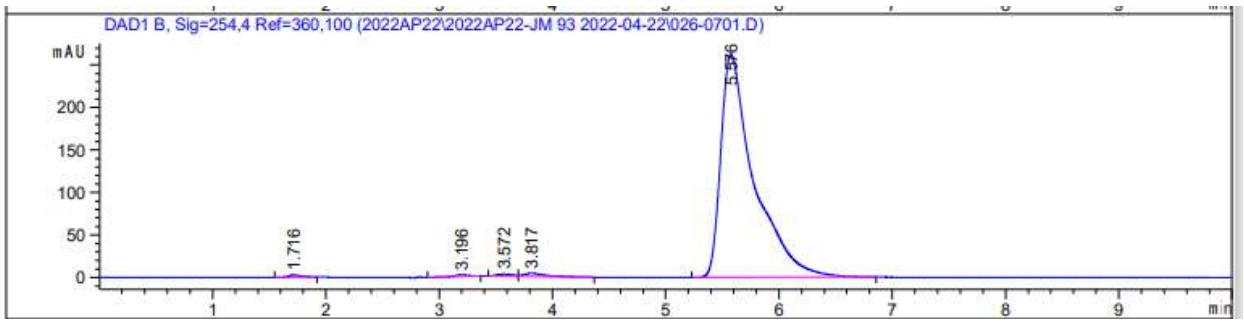

Figure 10: Chromatogram from Pure Carvone Bottom Layer

In order to determine the best steeping time and time to equilibrate, a series of experiments were set up to determine which had the best percent recovery. Of the four trials that worked, none showed promising results. Two trials, A and B, were set up for a 30-minute steep and a 24-hour equilibrate, trial C was a 24-hour steep and a one hour equilibrate, and trial D was a 24-hour steep with a 24-hour equilibrate. Table 11 shows these four trials and the percent yield determined from their top layer. These were all dried leaves that were crushed up and sifted and it was difficult to dry enough for all the experiments. Almost a full plant was necessary for one trial, and the percent yield was so small it was decided to use fresh leaves instead for the experiment.

Table 11. Testing of Different Steep and Equilibrate Times

Trial	Peak Area (mAU*s)	Concentration (mg/mL)	Top Volume (mL)	Amount of Carvone in Top (mg)	Starting mg	% Yield Carvone
A Top	28.69	0.00737	2.50	0.01842	1002.0	0.00184
B Top	31.79	0.00817	2.00	0.01633	992.70	0.00165
C Top	42.39	0.01089	1.50	0.01633	1005.2	0.00162
D Top	33.09	0.00850	2.00	0.01700	1075.7	0.00158

Imagining that the spearmint leaves would act like tea leaves, the longer the fresh leaves were allowed to steep, the more carvone and other materials should leach out of the leaves and into the solution. It seemed clear the amount of carvone found

was going to be extremely small, so a 24-hour steep time was chosen. It was determined though that there was no difference between the samples equilibrating for one hour or 24 hours as long as the temperature reached -20 °C, so one hour was chosen because it was a more manageable time.

According to Table 11, the yield of carvone from spearmint leaves (dried under ambient conditions) using the PIE method is a little under 0.002 %. This low result indicates that the drying process and the extraction method on dried leaves needs to be further investigated. According to Tables 12 and 13, the yield of carvone from fresh spearmint leaves is 0.004 %. In a separate study, we found that spearmint leaves dried to constant weight (40 ^{o}C, forced air oven) resulted in an 84 % loss of mass. Thus, the correction factor to convert the yield of carvone from fresh leaves to a dry weight basis is 6.25 (100 % fresh mass / (100 % – 84 %) dried mass). Assuming that the mass loss was due to loss of water allows us to calculate the yield of carvone from fresh spearmint leaves on a dry weight basis to approximately 0.024 %. This yield was also very low and indicates that the extraction method needs to be further investigated.

Two weeks of experiments were run with the fresh spearmint to determine the percent yield of carvone from the spearmint and the partition coefficient. The results from both weeks had some problems that have led to only one of those questions being able to have an answer. In the fresh experiment 1, all three trials had top layer chromatograms with a detectable carvone peak. The concentrations, amount of carvone in the top layers, and the percent yield were all able to be determined for these top layers as can be seen in Table 12. In the fresh experiment 2, only two of the three trials resulted in a detectable carvone peak from the HPLC. Trial one did not contain a

carvone peak and a percent yield could not be determined for this trial (Table 13). The average percent yield from the five trials with peaks was 0.0244% with a standard deviation of 0.00205 after the dried mass percent was taken into consideration and factored into the final calculations. A literature search identified a study that showed carvone was extracted in high concentrations from spearmint leaves.[10] The carvone percentage ranged from 59.74%-72.87% which leads to a yield of carvone between 0.82% and 1.34%.[10] Chrysargyris *et al.* studied the effect of the application of saline solutions to the leaves of hydroponically grown spearmint (*mentha spica*) on the yield of essential oil and the distribution of the major volatile components.[10] The essential oil was obtained by drying the leaves at 42 ^{o}C in a forced air oven followed by hydrodistillation. The yield of the essential oil from spearmint on a dry weight basis (mL oil/100 g dried leaves) was 1.5 – 2.0% showing carvone to be a major component in spearmint. The yield determined in this research is comparably small to literature value, but it is a great starting point for the study and hopefully better yields will be determined in future studies. More experiments need to be conducted to determine the true effectiveness of the PIE method for extracting carvone from spearmint. The PIE method was proven to separate carvone due to the large partition coefficient found in the pure carvone experiments, but the spearmint leaves did not release a large amount of carvone within the time of this experiment. Figure 11 gives one of the chromatograms for a top layer of the pure carvone analyzed on the HPLC.

[10] Chrysargyris, A.; Loupasai, S.; Petropouos, S. A.; Tzortzakis. N. Salinity and cation foliar application: Implications on essential oil yield and composition of hydroponically grown spearmint plants *Sci. Hortic.* **2019**, 256, 108581. https://doi.org/10.1016/j.scienta.2019.108581.

Table 12. Fresh Experiment 1 Top Layer Calculations

Fresh Experiment 1 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Top Volume (mL)	Amount of Carvone in Top (mg)	Starting mg	% Yield of Carvone	% Yield Adjusted for Moisture
Top 1	56.79	0.01459	2.50	0.03647	1000.9	0.00364	0.0228
Top 2	66.17	0.01699	2.30	0.03909	1002.3	0.00390	0.0244
Top 3	78.59	0.02019	2.20	0.04441	1001.1	0.00444	0.0278

Table 13. Fresh Experiment 2 Top Layer Calculation

Fresh Experiment 2 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Top Volume (mL)	Amount of Carvone in Top (mg)	Starting mg	% Yield of Carvone	% Yield Adjusted for Moisture
Top 1	N/A	N/A	2.50	N/A	1006.0	N/A	N/A
Top 2	57.03	0.01465	2.50	0.03662	1004.2	0.00365	0.0228
Top 3	51.26	0.01316	3.00	0.03949	1028.9	0.00384	0.0240

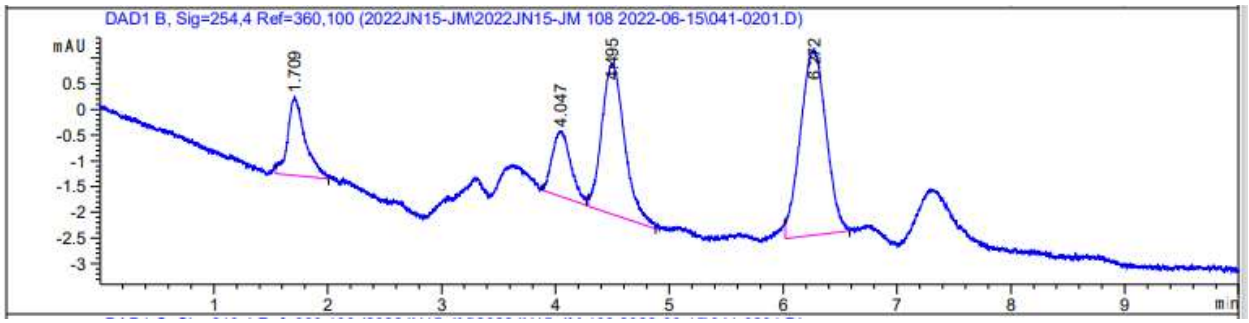

Figure 11. Chromatogram from Fresh Spearmint Leaves Top Layer

The bottom layers of both the fresh experiments seen in Tables 12 and 13 did not show peaks that were prominent enough to determine an accurate partition coefficient. All three trials from the fresh experiment 1 bottom layers lacked the carvone peak in the HPLC data. Without a peak area, the concentration and amount of carvone in the layer could not be determined (Table 14). The amount of carvone in the bottom layer was needed for determining the partition coefficient, so it could not be determined at all in this experiment.

Table 14. Fresh Experiment 1 Bottom Layer Calculations

Fresh Experiment 1 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Bottom Volume (mL)	Amount of Carvone in Bottom (mg)	Mass Balance (mg)	Partition Coefficient
Bottom 1	0	0	11.0	0	0.0365	N/A
Bottom 2	0	0	11.2	0	0.03909	N/A
Bottom 3	0	0	11.5	0	0.04441	N/A

In fresh experiment 2, there were small peaks for all three trials, but trial 1 was not able to be used because the top layer did not yield a peak so there was no data to compare it to. The peaks were found at a slightly lower retention time than where the carvone peak should have been, but due to the impurities of the natural product versus the pure carvone, it was determined the peaks were close enough to the expected peak range that they could be used for this data. Table 15 shows the calculations that were able to be completed. Two data points are not enough to differentiate from impurities and calculate a partition coefficient. Any carvone there should have shown a much higher affinity for the top layer and yielded a high partition coefficient. Figure 12 gives one of the chromatograms for a bottom layer of the pure carvone analyzed on the HPLC.

Table 15. Fresh Experiment 2 Bottom Layer Calculations

Fresh Experiment 2 Trial #	Peak Area (mAU*s)	Concentration (mg/mL)	Bottom Volume (mL)	Amount of Carvone in Bottom (mg)	Mass Balance (mg)	Partition Coefficient
Bottom 1	39.25	0.01008	11.0	0.1109	N/A	N/A
Bottom 2	36.04	0.00926	11.0	0.1018	0.1385	1.58
Bottom 3	35.10	0.00902	11.0	0.0992	0.1387	1.46

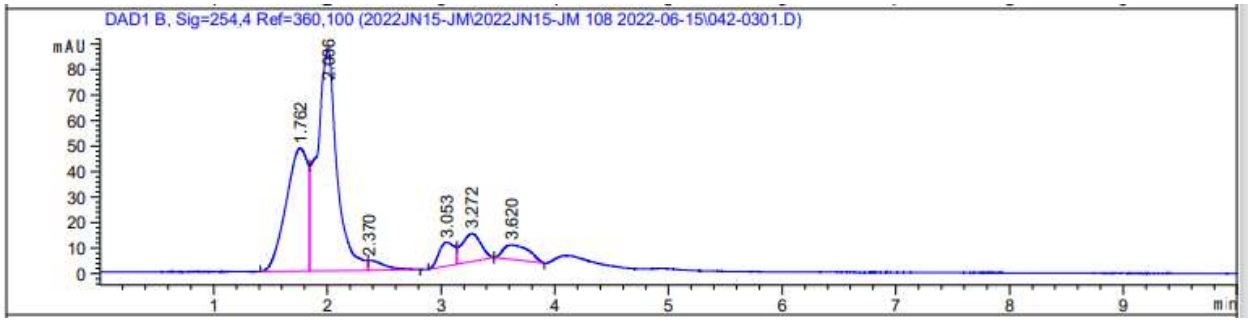

Figure 12. Chromatogram from Fresh Spearmint Leaves Bottom Layer

Of the three objectives, two proved to be successful. The first objective of identifying the partition coefficient of pure carvone via the PIE method yielded promising results. The only problems seen with this experiment would be the type of pure carvone being studied. From the introduction, it is known the (*R*)-(-) and (*S*)-(+) enantiomers of carvone act differently and elute at different times from the column in chromatography. While the (*S*)-(+) enantiomer was used as the pure carvone, the naturally occurring compound in spearmint would be the other enantiomer. It would be good to have an additional study completed using the other enantiomer as the pure compound to observe the differences in retention times and to use a different method of analysis which can differentiate between the two enantiomers.

The other two objectives of this research involved the percent yield, which was calculated, and partition coefficient of carvone extracted from fresh spearmint leaves. The carvone extracted out would have been in much lower concentrations than the pure carvone experiments. The (*R*)-(-) enantiomer (which is what should be present) should have been more easily detectable at the lower concentrations. When comparing the HPLC data peaks, the peak areas did not match the pure carvone peaks found earlier in

the research. This may be caused by contamination or a change in retention times from other components within the spearmint leaves.

Numerous other essential oils are present in spearmint and would have steeped out of the leaves during this process as well. The final solutions would have been full of impurities which could have led to unexpected retention times and peaks being overlooked in analysis. Further experimentation could be completed to try and eliminate some of those impurities present in the final solutions. An experiment completed by the University of Illinois at Chicago was able to successfully extract carvone from spearmint leaves using a steam distillation extraction method. Then, methylene chloride was used to separate the compounds in the product. Ethyl acetate would have also been a good solvent to isolate the carvone because it is a polar molecule with a low boiling point.[11] By distilling the solution, the water was evaporated off while the essential oils were extracted from the leaves which allowed for fresh leaves to be used, but the water was removed as if they were dry leaves.

The conditions of the experiment can also be adjusted to look for a more optimal experimental run. If the samples sat for too long at -20 °C, perhaps they began to freeze which would not allow for proper separation of the layers or for all the oils present to separate into their respective layers. Further experiments should also leave the leaves to steep for longer (maybe up to a week even) and at a higher temperature to allow all the oils to fully steep out of the leaves while evaporating off the excess water. This might provide a more accurate representation of the compounds present throughout the

[11] *University of Illinois* **2020/2021** "UIC Lab 3 Report." Studocu.com/en-us/document/university-of-illinois-at-chicago/organic-chemistry-laboratory-i/uic-lab-3-report/10205680 (accessed 2022-10-20).

whole leaf if it has time to digest. Additional solvents should also be tested. While the PIE method uses acetonitrile for the actual separation, additional solvents could be used initially to digest the spearmint and isolate the carvone. During this experiment, an acetonitrile and water mixture was used for everything including both the steeping process and the PIE method. Going forward, additional solvents such as ethyl acetate could be attempted to extract out the carvone, then acetonitrile can be used to separate the mixture.

The last area of improvement would be the use of fresh (wet) versus dried leaves. The dried leaves may have given a more accurate representation of the compounds present. Fresh leaves contain so much water, it was hard to determine a true yield because concentrations of the oils were so low. Once the dried mass and moisture content was accounted for, the yields increased enough to warrant further experimentation to try for a higher extraction. The dried leaves take a while to dry out though and there needs to be a consistency which is difficult when there are leaves of multiple sizes and ages. Dried leaves also end up needing to be crushed and sifted to make them more easily handled, but this opens them up to the introduction of other impurities. Overall, the experiments were completed in a timely manner and gave good data to build off even if that data went against the initial hypothesis.

Conclusion

There were three objectives for this study and not all of them resulted in complete findings. The determination of the partition coefficient of pure carvone was the first goal. This proved to be successful. Data showed the partition coefficient of pure (*S*)-(+)-carvone to be 120 with a standard deviation of 16.5 which indicated the PIE method was a successful extraction technique for pure carvone. The determination of percent yield of carvone in spearmint leaves was also successful. The average percent yield adjusted for the dried mass percent was determined to be 0.0244% with a standard deviation of 0.00205. This data was taken from five trials because the sixth trial did not even yield a carvone peak in the HPLC data. The determination of the partition coefficient of the raw material using the PIE method was not successful due to lack of bottom layer data. While the PIE method was proven to work on pure carvone, the results of this experiment remain inconclusive as to whether the PIE method can be used to efficiently extract carvone out of natural materials such as spearmint leaves in quantities large enough to make it useful. Further research needs to be conducted to get the essential oil out of the spearmint leaves in order to test the PIE method.

Acknowledgements

I would like to first and foremost thank Dr. Sowa for his guidance, expertise, and knowledge and for allowing me to work within his research group. Thank you also to Dr. Derick Jones who was often in the lab helping with experiments and brainstorming new ideas to try. Thank you to Governors State University for providing the research facilities and the opportunity to learn under such great professors. Thank you to the other members of the research group (especially Lena Moore). While we did not all work together on the same project, it was helpful to hear new ideas and experiments others were running and learn from people who had already been working with the PIE method before I did. Finally, thank you to my supportive family who was always pushing me to do my best and follow my dreams.